Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/03/2011
PI Name:	Hyers, Robert Ph.D.		
Project Title:	Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rwhyers@wpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	508-831-5759
Organization Name:	Worcester Polytechnic Institute		
PI Address 1:	100 Institute Road		
PI Address 2:			
PI Web Page:			
City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	2
Comments:	Formerly with University of Massachusetts at Amherst.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	09/01/2010	**End Date:**	08/31/2013
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Robert Hyers is U.S. Team Member on ESA-sponsored projects, "Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC)," "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)," and "Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL)."		
COI Name (Institution):			
Grant/Contract No.:	NNX10AR71G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Funding is for Dr. Hyers's role as Team Member on the subprojects described below, associated with ESA-sponsored main projects as noted.
Main project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany

Subprojects:

1. Transformation Kinetics-ISS. Task description: Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the MSL-EML to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection.

2. Thermophysical Properties. Task description: The density, viscosity, and surface tension of the flight samples will be measured in flight for a range of temperatures. Density will be determined by the video metrology, and surface tension and viscosity by the oscillating drop method. This WP includes determination of parameter modifications required for these measurements as well as collection, reduction, and analysis of the resulting data. The resulting data will support the whole range of theoretical and modeling efforts in this proposal.

3. Modeling of Macroconvection. Task description: Magnetohydrodynamic (MHD) model of macroconvection in the samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theorerical and modeling efforts. Collaboration on development of parameters for other work packages to enable the use of convective state as an experimental parameter.

Main project: Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany.

Subproject:

Modeling of macroconvection: Magnetohydrodynamic modelling of macroconvection in Ti-Zr-Ni samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theorerical and modeling efforts.

Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany

Subprojects:

Long Duration microgravity Experiments - ISS. Evaluation of microgravity experiments for fluid flow investigations.

MODELLING A: Thermodynamics, Phase Selection. Modelling of nucleation under various flow conditions for steel alloys.

MODELLING B: Non-contact thermophysical. Modeling of magnetohydrodynamic convection and shear in 1-g, microgravity for 4-6 alloys. Modeling of Marangoni convection in 1-g ESL drops for comparison with microgravity experiments. Modeling of temperature distribution in 1-g and microgravity samples, 4-6 alloys. Input into design of experimental parameters for 4-6 Alloys.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This project consists of modeling in support of three different flight experiments:
- PARSEC - Peritectic Alloy Rapid Solidification with EM Convection

- ThermoLAB - Thermophysical Properties of Liquid Metallic Alloys

- ICOPROSOL - Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order

In PARSEC, we are investigating the effect of fluid flow on the solidification path of certain alloys. Control of the solidification path would enable tailoring of the microstructure and properties of metal parts for specific applications.

In ThermoLAB, we are investigating the thermophysical properties of high-temperature materials. Many of these materials are used commercially. A better understanding of the physical properties will allow more efficient and more reliable production of metallic parts using these alloys. Other alloys are chosen because of scientific interest in their properties, which may illuminate fundamental physics related to the formation of bulk metallic glasses and quasicrystals.

In ICOPROSOL, we are investigating the nucleation and growth of quasicrystals, and the effect of atomic-scale order on the macroscopic properties of these alloys. This fundamental investigation may improve our ability to tailor the microstructure of metals for commercial applications.

Task Progress:	Highlights of Work Accomplished During Grant Period 2010-11. • ESA proposals passed science peer review and are approved 6/1/2010 - PARSEC - Peritectic Alloy Rapid Solidification with EM Convection - ThermoLAB - Thermophysical Properties of Liquid Metallic Alloys - ICOPROSOL - Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order • Hired Ms. Briana Tomboulian MS/PhD student at UMass to conduct flow modeling tasks. Ms. Tomboulian was selected for a NASA Science and Technology Research Fellowship, which will support her PhD Research. • Attended 3rd MSL-EML Working Group at ESA/ESTEC 11/15/2010 • Invited member ESA International Topical Team on Thermophysical Properties (ThermoLab-ISS kick-off meeting) at ESA/ESTEC 11/21/2010 • Rebuilt and validated magnetohydrodynamic model for laminar internal flow in electromagnetically levitated droplets in COMSOL Multiphysics. Turbulent flow model in progress and projected for completion in 2011. • Validated magnetic model with new MSL-EML flight coils against experiments by Dr. Georg Lohoefer, DLR, Cologne. • Rebuilt and validated Marangoni convection model with simple boundary conditions in COMSOL Multiphysics. Porting and validation of laser heating model is ongoing and projected for completion in 2011.
Bibliography Type:	Description: (Last Updated: 03/30/2026)
Abstracts for Journals and Proceedings	Hyers RW. "Turbulent Transition in EML Drops." Presented at the Thermolab Workshop, ESA ESTEC, Noordwijk, Netherlands, November 19, 2010. Presented at the Thermolab Workshop, ESA ESTEC, Noordwijk, Netherlands, November 19, 2010. , Nov-2010
Abstracts for Journals and Proceedings	Hyers RW. "Non-Contact Measurement of Thermophysical Properties." 6th Reactive Metals Workshop, Pasadena, CA, March 4-5, 2011. 6th Reactive Metals Workshop, Pasadena, CA, March 5, 2011. , Mar-2011
Abstracts for Journals and Proceedings	Hyers RW. "Fluid Flow Effects in Containerless Processing of Liquid Metals." ISPS-IV Satellite Conference: Workshop on Dynamics, Properties, and Solidification of Liquid Alloys (DyPSLA 2011), Cologne, Germany, 7-9 July 2011. ISPS-IV Satellite Conference: Workshop on Dynamics, Properties, and Solidification of Liquid Alloys (DyPSLA 2011), Cologne, Germany, 7-9 July 2011. , Jul-2011
Articles in Peer-reviewed Journals	Matson DM, Hyers RW, Volkmann T. "Peritectic alloy rapid solidification with electromagnetic convection." Journal of the Japanese Society of Microgravity Applications, 2010;27(4). , Dec-2010
Books/Book Chapters	Hyers RW. "Computer-Aided Experiments in Materials Processing." in "Solidification of containerless undercooled melts." Ed. D.M. Herlach, D.M. Matson. Wiley-VCH, in press, as of July 2011. Expected publication Fall 2011., Jul-2011
Papers from Meeting Proceedings	Hyers RW, Matson DM, Kelton KF, Holland-Moritz D, Volkmann T. "Fluid-Flow Effects on Phase Selection and Nucleation in Undercooled Liquid Metals" 4th International Symposium on Physical Sciences in Space, Bonn, Bad Godesberg, Germany,11-15 July 2011. Proceedings. 4th International Symposium on Physical Sciences in Space, Bonn, Bad Godesberg, Germany, 11-15 July 2011. , Jul-2011
Papers from Meeting Proceedings	Matson DM, Hyers RW. "Phase selection in the Mushy Zone: LODESTARS and ELFSTONE Programs." 4th International Symposium on Physical Sciences in Space, Bonn, Bad Godesberg, Germany,11-15 July 2011. Proceedings. 4th International Symposium on Physical Sciences in Space, Bonn, Bad Godesberg, Germany, 11-15 July 2011. p. 220-225. , , Jul-2011